Course Code : ECT 257 **QTSR/RW – 24 / 1570**

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

ANALOG CIRCUITS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.

(2) All questions carry marks as indicated against them.

(3) Assume suitable data and illustrate answer with sketches wherever necessary.

1. (a) Develop the expression for input and output resistance of current shut feedback amplifier. 8(CO1,2)

   (b) Write the steps to identify the feedback topology of the practical circuit. 2(CO1,2)

2. (a) Differentiate the various classes of operation of power amplifiers based on

   (a) Operating cycle

   (b) Position of Q point

   (c) Efficiency.

   Draw the input-output waveform for all power amplifier. 5(CO3)

   (b) Explain the working of complementary symmetry class B amplifier. What are its advantages and disadvantages ? 5(CO3)

3. (a) Explain the operation of LC tank Circuit. 2(CO2,3)

   (b) Explain the working of Colpitts oscillator. Develop the formula for the frequency of oscillations. 8(CO2,3)

4. (a) Illustrate the circuit diagram of Differential amplifier with swamping resistance. Explain the effect of swamping resistance on various parameters of Differential amplifier. 5(CO3)

   (b) Perform the DC analysis for Single input unbalanced output Differential amplifier. 5(CO3)

5. (a) Define :
   - (a) Slew Rate
   - (b) CMRR
   - (c) Supply voltage rejection ratio
   - (d) Input offset voltage
   - (e) Input bias current 5(CO4)

   (b) What is an Integrator ? What are its limitations of ideal Integrator ? How they are overcome in practical Integrator ? 5(CO4)

6. (a) Design 2nd order Butterworth High pass filter at a lower cut-off frequency 10KHZ. 5(CO5)

   (b) Design a 555 astable to operate at 8 KHz. Duty cycle is 50%. Assume suitable data, if necessary. 5(CO5)